**A project Report**
**On**

# "Zomato Restaurant Success Prediction"


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**Roll No : 08**

**Seat Number :**

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**MAHARASHTRA**

**2024-2025**

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# CERTIFICATE

**Roll No. 008** **Exam Seat No:**

This is to certify that the project entitled **"Zomato Restaurant Success Prediction"** , is bonafide work of **MAYUR KADAM** submitted in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE** in **DATA SCIENCE** from University of Mumba

**Internal Guide** **Head of Department**

Externals Examiner Date:

College Seal

# DECLARATION

I hereby- declare that the project entitled, "**Zomato Restaurant Success Prediction**" done at Viva College Virar West, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge, no one has submitted to any other university.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (DATA SCIENCE)** to be submitted as fifth semester project as part of our curriculum.

**MAYUR KADAM**

**(Signature)**

# ACKNOWLEDGEMENT

I am very grateful to our principal for providing us with an environment to complete my project successfully. I am deeply indebted to **Prof. SHWETA YANDE** Head of D.S Department, VIVA College who modelled us both technically and morally for achieving greater success in life.

I express our sincere thanks to all our lecturers, for their constant encouragement and support throughout our course, especially for the useful suggestions given during the course of the project period. I am grateful to my internal guide **Prof. JYOTI JADHAV** Lecturer, for being instrumental in the completion of our project with her complete guidance .

We would like to thank our parent for providing us with all their support and encouragement right from the project's budding stage to its current maturity .

Above all we would like to thank the almighty for giving us courage and energy to workday and night to make this project a grand success.

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# Abstract

The rapid expansion of the food delivery and restaurant industry has generated large volumes of structured and semi-structured data, creating new opportunities for data-driven decision-making. In a highly competitive environment where operational efficiency, pricing strategy, cuisine selection, and location play a crucial role in determining business success, predictive analytics can provide significant strategic advantages. This project focuses on developing a Machine Learning-based Restaurant Success Prediction System using a real-world restaurant dataset. The primary objective of this study is to analyze key restaurant attributes and build a predictive model capable of estimating whether a restaurant is likely to be successful or not based on operational and contextual features.

The dataset used in this project consists of restaurant-related information such as average price, average delivery time, cuisine types, and location details. Initially, the raw dataset (zomato_dataset.csv) was explored and cleaned to handle missing values, remove inconsistencies, and standardize categorical variables. After preprocessing, a refined dataset (cleaned_restaurant_data.csv) was created, which includes engineered features and binary encoded categorical variables suitable for machine learning algorithms. Feature engineering techniques were applied to transform cuisine types and locations into dummy variables, ensuring compatibility with classification models.

A supervised machine learning approach was adopted for prediction. Multiple classification algorithms were evaluated, and the most suitable model was selected based on performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score. The final trained model was saved and integrated into an interactive Streamlit web application (app.py), allowing users to input restaurant parameters and receive real-time predictions along with probability scores. Additionally, the application provides analytical dashboards for city-wise and cuisine-wise success analysis, enabling better interpretation of model outputs.

The results demonstrate that machine learning techniques can effectively capture the relationship between operational factors and restaurant performance. The developed system not only predicts success probability but also offers insights that can assist business owners in optimizing pricing strategies, reducing delivery times, and selecting suitable cuisine combinations for specific locations. This project highlights the practical implementation of end-to-end machine learning workflow, from data preprocessing to deployment, and showcases the potential of predictive analytics in supporting strategic decision-making in the restaurant industry.

# Chapter 1 – Introduction

## 1.1 Background of the Online Food Delivery Industry

The online food delivery industry has emerged as one of the fastest-growing segments of the global service economy, driven by rapid digitalization, urbanization, and changing consumer lifestyles. Traditionally, the food industry relied heavily on dine-in services and telephone-based takeaway orders. However, with the widespread adoption of smartphones, affordable internet access, and secure digital payment systems, customers increasingly prefer ordering food through mobile applications. This shift has transformed the operational structure of restaurants and introduced technology-driven platforms that act as intermediaries between customers and food service providers.

Online food delivery platforms operate as digital marketplaces that connect restaurants, customers, and delivery partners in a single integrated system. These platforms allow customers to browse restaurant menus, compare prices, read reviews, check ratings, and place orders instantly. From the restaurant's perspective, online platforms provide extended market reach, digital visibility, and access to performance analytics. The convenience and transparency offered by these platforms have significantly increased customer expectations regarding pricing, delivery time, food quality, and service efficiency.

A defining characteristic of the online food delivery industry is its data-centric nature. Every customer interaction generates valuable data, including ordering frequency, preferred cuisines, spending patterns, delivery performance, and geographic demand trends. This continuous generation of structured data creates opportunities for applying advanced analytics and machine learning techniques. Restaurants and platforms can analyze this data to optimize pricing strategies, improve delivery logistics, identify high-demand cuisines, and enhance overall customer satisfaction.

In India and many other countries, companies like Zomato and Swiggy have played a pivotal role in accelerating the growth of the online food delivery ecosystem. These platforms have not only digitized restaurant operations but have also introduced competition-driven environments where operational efficiency directly impacts business success. Restaurants must carefully manage factors such as average price, delivery time, cuisine selection, and location suitability to remain competitive within the platform ecosystem.

Despite the opportunities provided by digital platforms, the online food delivery industry also presents significant challenges. High competition, commission-based revenue models, fluctuating customer demand, and operational inefficiencies can affect restaurant profitability. Therefore, understanding the determinants of restaurant success has become essential for sustainable growth. Predictive analytics offers a powerful solution by identifying patterns within historical restaurant data and estimating the likelihood of success based on measurable features.

In this project, the online food delivery industry forms the foundational context for developing a Restaurant Success Prediction System. By analyzing structured restaurant data and applying machine learning techniques, the study aims to predict restaurant success probability and provide data-driven insights. This approach highlights how predictive modeling can support strategic decision-making in the modern, technology-driven food delivery ecosystem.

Another challenge encountered during development was feature imbalance. Certain cuisines appeared very frequently in the dataset while others appeared rarely. This caused the machine learning model to favor dominant cuisines during training. To address this issue, careful preprocessing and evaluation strategies were required to ensure fair learning across all categories.

Additionally, handling multi-label categorical features required careful transformation. A single restaurant could offer multiple cuisines, which meant the dataset did not follow a simple one-category-per-row structure. Converting such information into a machine-readable format required iterative splitting, trimming, and encoding operations. This significantly increased preprocessing complexity.

The project also required careful verification of prediction outputs. A model with high accuracy does not always guarantee practical usefulness. Therefore, evaluation had to focus not only on numerical metrics but also on logical consistency of predictions. For example, unrealistic predictions such as extremely high-priced restaurants in low-demand areas being classified as successful required manual validation and interpretation.

Deployment compatibility was another technical challenge. Machine learning models often behave differently when moved from notebook environment to application environment. Ensuring consistent preprocessing, column order, and encoding between training and deployment required saving column structures and validating inputs before prediction.

## 1.2 The Challenge Faced During Project Development

The development of the Restaurant Success Prediction System involved multiple technical and analytical challenges at different stages of the machine learning pipeline. Since the project is based on real-world restaurant data obtained from an online food delivery ecosystem, the dataset initially contained inconsistencies, missing values, and categorical complexity that required careful preprocessing before model training. Handling these challenges was essential to ensure the reliability and accuracy of the predictive system.

One of the primary challenges encountered during the project was data cleaning and preprocessing. The raw dataset (zomato_dataset.csv) contained missing entries in important features such as average price and delivery time. Additionally, categorical variables such as cuisine types and locations were represented in text format, often containing multiple values in a single field. Converting these complex categorical attributes into structured numerical features suitable for machine learning required extensive feature engineering. Dummy variable creation for cuisines and locations significantly increased the dimensionality of the dataset, making model training computationally intensive.

Another major challenge was defining the target variable “Success.” Since restaurant success can be interpreted in various ways (such as ratings, order frequency, or revenue), selecting a clear and consistent success criterion was critical. The definition had to align with measurable data attributes available in the cleaned dataset (cleaned_restaurant_data.csv). Any inconsistency in defining success could lead to biased or misleading model predictions.

Model selection and performance evaluation also presented challenges. Different classification algorithms perform differently depending on data structure and feature distribution. Logistic Regression provided interpretability but sometimes lacked predictive strength, while tree-based models such as Random Forest offered better accuracy but required parameter tuning. Selecting the most appropriate model required systematic experimentation and evaluation using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

Another challenge was ensuring consistency between training and deployment. The feature columns used during model training needed to exactly match the input structure expected by the Streamlit application (app.py). To address this, the list of model input columns was saved using model_columns.pkl. Any mismatch between training features and application inputs could cause runtime errors or incorrect predictions.

Finally, integrating the trained model into a user-friendly Streamlit web application required careful design to ensure usability and reliability. The application had to dynamically generate input features based on selected cuisines and locations, calculate prediction probabilities, and display results in an interpretable format. Balancing technical complexity with simplicity of user interface design was an important consideration during deployment.

Overall, the project involved challenges related to data preprocessing, feature engineering, model selection, evaluation, and deployment. Successfully addressing these issues strengthened the robustness of the Restaurant Success Prediction System and enhanced its practical applicability in the online food delivery industry.

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## 1.3 Growth of Machine Learning in Prediction Tasks

The rapid advancement of technology and the exponential growth of digital data have significantly accelerated the development and adoption of Machine Learning (ML) in prediction tasks across various industries. Machine learning, a subset of artificial intelligence, enables systems to automatically learn patterns from historical data and make informed predictions without being explicitly programmed. Over the past decade, the availability of large datasets, increased computational power, and improved algorithms have transformed machine learning from a research concept into a practical business tool used worldwide.

Prediction tasks are one of the most important applications of machine learning. Organizations use predictive models to forecast outcomes such as customer behavior, product demand, financial risk, disease diagnosis, fraud detection, and market trends. Instead of relying solely on traditional statistical techniques or manual decision-making, machine learning algorithms analyze complex relationships among multiple variables to identify hidden patterns. These models can handle high-dimensional data, nonlinear relationships, and large-scale datasets more efficiently than conventional analytical methods.

The growth of machine learning in prediction tasks is strongly connected to the expansion of data-driven industries such as e-commerce, finance, healthcare, transportation, and online services. Digital platforms continuously generate structured and unstructured data from user interactions, transactions, and operational activities. This data provides valuable insights when analyzed using predictive algorithms. For example, recommendation systems predict customer preferences, credit scoring models predict loan default risk, and demand forecasting models estimate future sales. These predictive capabilities help organizations make strategic decisions, reduce risks, and optimize resources.

In the context of the online food delivery industry, machine learning plays a crucial role in predicting customer preferences, estimating delivery times, detecting fraud, optimizing pricing strategies, and analyzing restaurant performance. Platforms use predictive analytics to improve customer satisfaction and operational efficiency. Restaurants can leverage these insights to adjust their pricing, manage delivery logistics, and choose appropriate cuisine combinations based on market demand. As competition increases in digital marketplaces, predictive modeling becomes an essential tool for sustaining business growth.

The advancement of machine learning algorithms such as Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, and Neural Networks has further enhanced predictive performance. These models can process complex datasets containing both numerical and categorical features. Additionally, evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC enable systematic comparison and validation of models. With the integration of cloud computing and open-source libraries like Scikit-learn and XGBoost, machine learning solutions have become more accessible and scalable.

In this project, the growth of machine learning in prediction tasks provides the foundation for developing a Restaurant Success Prediction System. By analyzing historical restaurant data and applying supervised classification algorithms, the model estimates the probability of success based on operational features such as price, delivery time, cuisine type, and location. This demonstrates how modern predictive techniques can transform raw business data into actionable intelligence. The increasing reliance on machine learning across industries highlights its importance as a strategic tool for data-driven decision-making in the digital era.

Machine learning has evolved from rule-based systems to adaptive learning systems capable of improving performance over time. Earlier decision-making systems relied on predefined conditions, whereas modern models learn relationships automatically from historical data. This shift allows organizations to respond to dynamic environments rather than static conditions.

In business environments, predictive modeling reduces uncertainty. Instead of reacting to failures, organizations can proactively identify risks and opportunities. For example, predicting low performance before launching a restaurant allows owners to modify strategies such as pricing or menu offerings. This reduces financial loss and improves planning efficiency.

The increasing accessibility of open-source tools has democratized machine learning adoption. Libraries such as Scikit-learn allow students and small businesses to implement predictive systems without large infrastructure investment. This has made machine learning a practical tool rather than a specialized research domain.

Another important development is explainable machine learning. Businesses not only need predictions but also need to understand why predictions occur. Feature importance analysis helps stakeholders interpret model behavior and build trust in automated decision systems. This is especially important in business planning scenarios where decisions involve financial investments.

## 1.4 Scope of the Study

The scope of this study is centered on the development and evaluation of a machine learning–based Restaurant Success Prediction System using structured restaurant data derived from the online food delivery ecosystem. The project focuses specifically on analyzing operational and contextual attributes such as average price, average delivery time, cuisine types, and geographic location to determine their impact on restaurant success. By limiting the analysis to measurable and structured features available in the dataset, the study ensures that the predictive model remains data-driven, objective, and practically applicable.

This study covers the complete machine learning pipeline, beginning with data preprocessing and feature engineering. The raw dataset (zomato_dataset.csv) is cleaned to remove inconsistencies, handle missing values, and standardize categorical variables. A refined dataset (cleaned_restaurant_data.csv) is then created, containing numerical representations of categorical features through encoding techniques such as dummy variable creation. The scope includes preparing the dataset in a format suitable for supervised classification models and ensuring that the input features align with the requirements of the final deployed application.

The project also encompasses model development and evaluation. Multiple machine learning algorithms are considered to identify the most effective approach for predicting restaurant success. The study evaluates model performance using standard classification metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. However, the scope does not extend to advanced deep learning architectures or real-time streaming data integration, as the focus is on building a robust and interpretable predictive model using structured tabular data.

Another important component within the scope is the deployment of the trained model into a user-friendly web application using Streamlit (app.py). The application allows users to input restaurant characteristics and receive a predicted success probability. Additionally, it provides analytical dashboards for city-wise and cuisine-wise insights. This integration ensures that the study moves beyond theoretical modeling and demonstrates practical implementation in a real-world scenario.

It is important to note that the scope of this study is limited to predictive analysis based on historical data. External factors such as sudden market disruptions, seasonal demand variations, government regulations, or macroeconomic changes are not incorporated into the model. Furthermore, the success definition is based solely on the attributes available in the dataset and does not include real-time financial metrics such as revenue or profit margins.

In summary, the scope of this study includes data preprocessing, feature engineering, model training, evaluation, and deployment of a predictive system for restaurant success within the online food delivery domain. By focusing on measurable operational factors and structured data, the project demonstrates how machine learning can support data-driven decision-making while maintaining clear boundaries to ensure feasibility and accuracy.

The study is limited to structured numerical and categorical attributes available in the dataset and does not include external behavioral or psychological factors influencing consumer decisions. Customer preferences may also depend on branding, marketing campaigns, or social media popularity, which are outside the scope of this model.

Additionally, the model is designed for prediction rather than optimization. It predicts the likelihood of success but does not automatically modify inputs to generate the best configuration. However, users can manually test different input combinations to identify favorable conditions.

The scope also focuses on classification rather than forecasting. The system predicts success category based on existing conditions but does not estimate future revenue values or growth rates. Future systems may incorporate time-series forecasting for long-term performance estimation.

## 1.5 Objectives of the Study

The main objectives of this project are as follows:

1. **To analyze restaurant data** collected from the online food delivery ecosystem and understand key operational features affecting performance.
2. **To clean and preprocess the raw dataset (zomato_dataset.csv)** by handling missing values, removing inconsistencies, and preparing structured data for analysis.
3. **To perform feature engineering** by converting categorical variables such as cuisines and locations into numerical formats suitable for machine learning models.
4. **To define and construct a target variable (Success)** based on measurable criteria available in the dataset.
5. **To develop a supervised machine learning classification model** capable of predicting whether a restaurant is likely to be successful or not.
6. **To compare multiple machine learning algorithms** and select the best-performing model using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
7. **To evaluate model performance systematically** using confusion matrix analysis and other validation techniques.
8. **To deploy the trained model into a Streamlit web application (app.py)** for real-time prediction and user interaction.
9. **To provide analytical insights** such as city-wise and cuisine-wise success analysis through interactive dashboards.
10. **To demonstrate the practical application of machine learning** in supporting strategic decision-making within the online food delivery industry.
11. From an economic perspective, predictive analytics reduces business risk. New restaurants often require significant investment in infrastructure, staff, and marketing. Predicting performance beforehand allows investors to make informed decisions and allocate resources efficiently.
12. From a technological perspective, the study demonstrates integration of analytics with application deployment. Many academic projects end at model training, but practical systems require user interaction and usability. This project bridges that gap by converting a predictive model into a usable decision-support tool.
13. From a learning perspective, the project strengthens understanding of real-world data challenges. Unlike textbook datasets, real datasets contain noise, inconsistencies, and missing values. Handling such issues develops practical analytical skills necessary in industry environments.

## 1.6 Importance of the Study

The importance of this study lies in its ability to bridge the gap between traditional business decision-making and data-driven predictive analytics within the online food delivery industry. In a highly competitive digital marketplace, restaurant owners must make strategic decisions regarding pricing, cuisine selection, delivery efficiency, and location targeting. These decisions significantly impact business performance and long-term sustainability. However, relying solely on intuition or experience may not always yield optimal results. Therefore, applying machine learning techniques to analyze historical restaurant data provides a more systematic and evidence-based approach to decision-making.

This study is important because it demonstrates how structured data generated by online food delivery platforms can be transformed into actionable insights. The dataset used in this project contains measurable attributes such as average price, delivery time, cuisine types, and geographic location. By analyzing these factors through machine learning algorithms, the project identifies patterns that influence restaurant success. This helps stakeholders understand which operational parameters contribute positively or negatively to performance.

Another significant aspect of this study is its practical implementation. Unlike purely theoretical research, this project develops a complete predictive system integrated into a Streamlit web application. This enables real-time prediction of restaurant success probability based on user inputs. Such an application can assist restaurant owners, investors, and business analysts in evaluating potential restaurant configurations before making financial investments. This reduces risk and improves strategic planning.

The study is also important from an academic perspective. It provides a clear example of the end-to-end machine learning lifecycle, including data preprocessing, feature engineering, model training, evaluation, and deployment. Students and researchers can use this project as a reference for understanding how machine learning models are developed and implemented in real-world business scenarios. It highlights the practical relevance of classification algorithms and performance metrics in solving industry-related problems.

Furthermore, in the broader context of the digital economy, predictive analytics has become an essential tool for sustaining competitive advantage. As industries continue to generate large volumes of data, organizations that effectively analyze and utilize this data will outperform competitors. This project reinforces the importance of machine learning as a strategic business intelligence tool within the modern online food delivery ecosystem.

In conclusion, the importance of this study lies in its contribution to data-driven decision-making, practical machine learning implementation, and strategic analysis within the restaurant and food delivery industry. It demonstrates how predictive modeling can transform raw operational data into meaningful business insights, ultimately supporting growth and sustainability in a competitive digital marketplace.

From an economic perspective, predictive analytics reduces business risk. New restaurants often require significant investment in infrastructure, staff, and marketing. Predicting performance beforehand allows investors to make informed decisions and allocate resources efficiently.

From a technological perspective, the study demonstrates integration of analytics with application deployment. Many academic projects end at model training, but practical systems require user interaction and usability. This project bridges that gap by converting a predictive model into a usable decision-support tool.

From a learning perspective, the project strengthens understanding of real-world data challenges. Unlike textbook datasets, real datasets contain noise, inconsistencies, and missing values. Handling such issues develops practical analytical skills necessary in industry environments.

# Chapter 2 – Problem Statement

## 2.1 Overview of the Problem

The online food delivery industry has evolved into a highly competitive and data-intensive ecosystem where restaurants compete on multiple measurable parameters. Customers now rely on digital platforms to compare restaurants based on pricing, cuisine offerings, delivery time, ratings, and location convenience before making purchasing decisions. This increased transparency has intensified competition, making it difficult for restaurants to predict their performance outcomes. While restaurants generate large amounts of operational data, most business decisions are still made based on intuition, limited analysis, or past experience rather than systematic data-driven evaluation.

The primary problem addressed in this project is the absence of a structured and predictive system that can estimate restaurant success using available operational features. Restaurant owners and investors often face uncertainty when launching new restaurants or modifying business strategies. Without predictive analytics, decisions such as setting average price, choosing cuisine combinations, or selecting a target location involve significant financial risk. Therefore, there is a need to develop a machine learning model capable of analyzing historical restaurant data and predicting the likelihood of success in a measurable and reliable manner.

In the digital marketplace, restaurant performance is influenced by a combination of operational efficiency and consumer perception. Customers evaluate multiple restaurants simultaneously before making a decision, which makes competition multidimensional rather than location-based alone. Unlike traditional markets where competition was limited to nearby establishments, online platforms create a virtual environment where all restaurants within a delivery radius compete directly with each other.

Another aspect of the problem is uncertainty in business planning. When entrepreneurs open a new restaurant, they must decide pricing, cuisine selection, and delivery strategies without clear predictive guidance. These decisions often rely on market observation or imitation of competitors, which does not guarantee success. A predictive system can convert historical patterns into measurable probabilities, reducing guesswork.

The availability of historical restaurant data provides an opportunity to transform descriptive analysis into predictive intelligence. However, raw data alone does not automatically produce insights. It requires structured modeling techniques capable of learning relationships between operational variables and outcomes. Therefore, the problem is not only the absence of information but the absence of a structured analytical mechanism to utilize the information effectively.

## 2.2 Complexity in Defining Restaurant Success

One of the fundamental challenges in this project lies in defining what constitutes “restaurant success.” Success in the restaurant industry can be interpreted in multiple ways, including high revenue generation, strong customer ratings, large order volumes, repeat customer retention, or long-term sustainability. However, not all these indicators are directly available in the dataset used for this study. This creates a conceptual challenge: the success variable must be clearly defined using measurable and consistent attributes present in the data.

An inconsistent or ambiguous definition of success may lead to incorrect labeling of training data, which directly affects model performance. Therefore, it is essential to establish a standardized and quantifiable criterion for success based on operational features available in the cleaned dataset. This ensures that the machine learning model learns patterns that are aligned with the project’s objective and produces reliable classification outcomes.

Success in digital food platforms is multi-dimensional and dynamic rather than static. A restaurant performing well today may decline tomorrow due to changes in consumer trends, competitor entry, or pricing shifts. Therefore, defining success requires choosing a measurable indicator that remains consistent across the dataset while still representing real performance.

Another complication is the difference between popularity and profitability. A restaurant with high orders may still be unprofitable due to high operational costs, while a moderately popular restaurant with optimized pricing may generate higher profit. Since financial records were not available in the dataset, the project required selecting an observable proxy variable representing performance.

Moreover, ratings alone may not accurately reflect success because they can be influenced by subjective opinions, temporary service issues, or promotional campaigns. Therefore, the project needed a definition that balances objectivity and practicality using structured attributes. Establishing this definition was crucial because machine learning models learn exactly what they are trained to predict. An unclear definition would produce misleading outcomes.

## 2.3 Limitations of Traditional Decision-Making Approaches

Traditional business decision-making in the restaurant industry often relies on intuition, personal experience, or basic market observation. While experience can provide valuable insights, it may not capture complex interactions between multiple variables simultaneously. For example, a higher average price may reduce demand in some locations but may not affect performance in premium urban areas. Similarly, offering multiple cuisines might increase customer reach but could also increase operational complexity.

Human decision-making has limitations in processing high-dimensional data and identifying nonlinear relationships between variables. As the number of influencing factors increases, it becomes increasingly difficult to manually analyze patterns and predict outcomes accurately.

Therefore, relying solely on traditional approaches may result in suboptimal decisions, increased financial risk, and inconsistent performance.

Human decision-making is often influenced by cognitive biases. Business owners may overestimate the importance of certain factors based on personal experience while ignoring others supported by data. For example, a restaurant owner may believe cuisine quality alone determines success, while actual performance may depend more on delivery time or pricing balance.

Another limitation is inability to process large datasets simultaneously. When multiple variables interact with each other, manual analysis becomes impractical. A restaurant may succeed only when several conditions occur together, such as moderate pricing combined with fast delivery in a high-demand location. Identifying such complex patterns manually is extremely difficult.

Traditional surveys and market research methods also have constraints. They capture opinions at a particular time but cannot continuously adapt to changing trends. In contrast, machine learning models update understanding based on historical patterns and can generalize to new unseen conditions. This makes predictive systems more reliable for long-term strategic planning.

## 2.4 Data-Related and Technical Challenges

The dataset used in this project contains both numerical features (such as average price and delivery time) and categorical features (such as cuisine types and locations). Categorical attributes often contain multiple values within a single record, especially in the case of cuisine combinations. Converting these attributes into numerical representations requires encoding techniques such as dummy variable creation.

However, encoding multiple cuisine types and locations significantly increases the number of features, leading to high-dimensional data. High dimensionality can cause computational challenges and increase the risk of overfitting during model training. Additionally, the raw dataset may contain missing values, inconsistent formatting, or duplicate entries, all of which must be addressed during preprocessing. Proper data cleaning and transformation are crucial to ensure model accuracy and prevent biased predictions.

High-dimensional data introduces another challenge known as the “curse of dimensionality.” When the number of features increases significantly due to encoding, the distance between data points becomes less meaningful for some algorithms. This may reduce model performance if not handled properly. Tree-based algorithms were considered partly because they handle high-dimensional categorical data more effectively.

Data imbalance also required attention. If one class appears more frequently than another, the model may become biased toward predicting the majority class. Therefore, evaluation metrics beyond accuracy were necessary to confirm balanced performance.

Additionally, preprocessing must be identical during both training and prediction. Even minor differences in column ordering or missing feature columns can cause incorrect predictions. To avoid this issue, feature columns were saved and reused in deployment. This ensures that the input structure matches the model expectations exactly.

Another technical challenge involved maintaining reproducibility. Machine learning models may produce slightly different results each time due to randomness. Fixing random_state ensures consistent results during evaluation and demonstration.

## 2.5 Need for a Practical and Deployable Predictive System

Beyond building a predictive model, the problem also involves creating a system that can be practically used by stakeholders. A theoretical model without deployment has limited real-world value. Restaurant owners require a user-friendly interface where they can input parameters such as price, delivery time, cuisine type, and location to obtain a predicted success probability.

Therefore, the problem extends beyond model development to system integration and deployment. The model must be integrated into a web-based application that ensures consistency between training features and real-time inputs. Additionally, the system should provide interpretable outputs such as probability scores and analytical dashboards for better decision-making. Ensuring usability, reliability, and scalability is a key aspect of solving the overall problem.

A predictive model has practical value only when accessible to decision makers. Business owners may not understand programming environments or data analysis notebooks. Therefore, the system must translate technical outputs into simple, interpretable results such as probability scores or categorical labels.

The application must also prevent invalid inputs. For instance, entering unrealistic values like negative delivery time should not generate predictions. Input validation mechanisms are necessary to maintain reliability.

Another important requirement is response speed. Predictions should be generated instantly so users can experiment with multiple configurations. Fast response encourages exploratory analysis where users adjust parameters and observe how predictions change.

Visualization also plays a role in usability. Graphical outputs such as success probability indicators and analysis dashboards improve comprehension compared to raw numerical outputs. Therefore, the deployable system must combine analytics with user interface design principles.

## 2.6 Final Problem Definition

In summary, the core problem addressed in this project is the development of a robust machine learning classification system that predicts restaurant success using structured operational data from the online food delivery ecosystem. The system must address challenges related to data preprocessing, feature engineering, success definition, model selection, performance evaluation, and deployment. By solving these challenges, the project aims to transform raw restaurant data into predictive intelligence that reduces uncertainty, supports strategic planning, and enhances decision-making within a competitive digital marketplace.

The final problem can therefore be viewed as a predictive decision-support challenge rather than a purely academic classification task. The objective is not only to classify restaurants but also to assist stakeholders in evaluating strategic choices before implementation.

The system must satisfy three requirements simultaneously:

1. Analytical reliability – predictions must be statistically meaningful
2. Practical usability – outputs must be understandable to users
3. Technical consistency – training and deployment must produce identical behavior

By addressing these requirements, the project converts historical restaurant data into actionable intelligence. The final deliverable becomes a tool capable of assisting planning decisions rather than merely demonstrating algorithm performance.

# Chapter 3 – Data Description

## 3.1 Data Description

The dataset used in this project is obtained from the online food delivery ecosystem and contains structured restaurant-related information. During the development of this project, two versions of the dataset were utilized:

1. **zomato_dataset.csv** – The original raw dataset
2. **cleaned_restaurant_data.csv** – The processed and cleaned dataset created during this project

The file zomato_dataset.csv represents the original dataset containing raw restaurant data. This dataset included inconsistencies such as missing values, duplicate records, mixed categorical formatting, and complex text-based cuisine and location attributes. Since raw datasets are often not directly suitable for machine learning models, several preprocessing steps were applied to improve data quality and structure.

The dataset cleaned_restaurant_data.csv was generated after performing data cleaning, transformation, and feature engineering. This cleaned dataset includes properly formatted numerical features and encoded categorical variables using dummy variable techniques. All irrelevant columns were removed, missing values were handled appropriately, and categorical attributes such as cuisines and locations were converted into binary indicator columns.

In this project report, both datasets are mentioned to clearly demonstrate the data preprocessing workflow. However, the machine learning model was trained exclusively on the cleaned dataset (cleaned_restaurant_data.csv), as it represents the final structured form of the data suitable for classification tasks.

Each row in the cleaned dataset corresponds to an individual restaurant, and each column represents a specific feature used for prediction. The final dataset structure ensures compatibility with machine learning algorithms and supports accurate success prediction.

In the Jupyter Notebook write the code to inserted the dataset into the variable :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MultiLabelBinarizer

df = pd.read_csv("zomato_dataset.csv")
```

Display the Original dataset of top 5 rows :

```
zomato.head()
```

| | Restaurant Name | Rating | Cuisine | Average Price | Average Delivery Time | Safety Measure | Location |
|---|---|---|---|---|---|---|---|
| 0 | Campus Bakers | 4.3 | Bakery, Fast Food, Pizza, Sandwich, Burger | ₹50 for one | 36 min | Restaurant partner follows WHO protocol | Agra |
| 1 | Mama Chicken Mama Franky House | 4 | North Indian, Mughlai, Rolls, Burger, Momos | ₹50 for one | 22 min | Follows all Max Safety measures to ensure your... | Agra |
| 2 | GMB - Gopika Sweets & Restaurant | 4.2 | North Indian, South Indian, Chinese, Fast Food... | ₹50 for one | 27 min | Follows all Max Safety measures to ensure your... | Agra |
| 3 | Shree Bankey Bihari Misthan Bhandar | 4.2 | Mithai, Street Food, South Indian, Chinese, Ic... | ₹50 for one | 28 min | Restaurant partner follows WHO protocol | Agra |
| 4 | Burger King | 4.2 | Burger, Fast Food, Beverages | ₹50 for one | 26 min | Follows all Max Safety measures to ensure your... | Agra |

In the second dataset is (cleaned_restaurant_data.csv) is created thought Original dataset .

It is created for Analysis purpose it is create by this code:

```
df.to_csv("cleaned_restaurant_data.csv", index=False)
```

Real-world datasets rarely exist in a ready-to-use analytical format. The original dataset used in this project represents data collected from a real platform environment, meaning it was created primarily for operational usage rather than analytical purposes. As a result, many attributes were designed for human readability instead of computational processing. This required careful transformation before the data could be used in machine learning algorithms.

Another important aspect of the dataset is heterogeneity. The dataset contains numerical, categorical, and multi-value text attributes within the same table. Machine learning algorithms require structured numerical input, therefore a transformation pipeline was necessary to convert diverse information into uniform numerical representation. This conversion process ensures that all restaurants can be compared on a common analytical scale.

The dataset also reflects behavioral patterns of customers indirectly. Although the data does not explicitly include customer behavior fields, operational attributes such as delivery time and pricing act as proxies for customer satisfaction and demand. Thus, the dataset provides indirect insight into consumer preferences and business performance.

Data quality validation was an important step before modeling. Each column was inspected for anomalies such as unexpected values, inconsistent formatting, and unrealistic ranges. These checks prevent the model from learning incorrect patterns that originate from data errors rather than actual business relationships.

## 3.2 Input Features (Independent Variables)

The selection of input features plays a critical role in predictive modeling. Features must represent meaningful business characteristics rather than random attributes. Each selected feature in this project corresponds to a decision a restaurant owner can control, such as pricing or menu type. This ensures that predictions remain actionable rather than theoretical.

Feature interpretability was also considered. Highly complex features may improve accuracy but reduce usability. Therefore, the project prioritized understandable features that stakeholders can easily modify in real scenarios. For example, average delivery time is a measurable operational parameter that management can improve through logistics optimization.

Feature independence was evaluated to reduce redundancy. Highly correlated features may cause certain models to overweight specific information. By maintaining relevant but non-duplicate features, the model learns balanced relationships and produces stable predictions.

### 3.2.1 Numerical Features

Numerical attributes are particularly valuable because they preserve magnitude relationships. Unlike categorical features, numerical values contain ordering and distance information. For instance, a delivery time of 20 minutes is objectively better than 40 minutes, and the model can learn this directly.

Normalization was considered but not strictly necessary for tree-based models such as Random Forest. However, numerical ranges were still verified to ensure there were no extreme outliers that could distort learning. Extremely unrealistic values were treated during preprocessing to maintain data reliability.

The relationship between price and success is not strictly linear. Very low prices may indicate poor quality perception, while very high prices may reduce affordability. Therefore, the model must learn optimal ranges rather than simple increasing or decreasing trends.

• **Average Price**
This feature represents the average cost of food items offered by the restaurant. Pricing plays a significant role in customer decision-making and directly impacts demand and profitability.

• **Average Delivery Time**
This feature indicates the estimated time required to deliver food to customers. Faster delivery generally improves customer satisfaction and increases the likelihood of repeat orders.

These numerical features were directly used in the model without requiring encoding.

### 3.2.2 Categorical Features (Encoded)

Categorical features originally existed in textual form. Machine learning algorithms cannot directly interpret text labels because they lack mathematical meaning. Encoding transforms qualitative descriptions into quantitative indicators.

One-hot encoding was selected because it prevents false ranking relationships. If label encoding were used, the model might assume that one cuisine type is "greater" than another numerically, which has no logical interpretation. Binary indicator columns instead represent presence or absence of characteristics.

The increase in columns after encoding expands the feature space, allowing the model to analyze specific cuisine combinations. This enables detection of patterns such as certain cuisine pairs performing better in particular locations.

The original dataset contained categorical variables such as cuisine types and location names. Since machine learning models require numerical inputs, these variables were converted into binary (0/1) encoded columns using one-hot encoding.

• **Cuisine Type Indicators**
Each cuisine (e.g., North Indian, Chinese, Fast Food, etc.) was converted into a separate binary column.

- 1 indicates the restaurant offers that cuisine
- 0 indicates absence

This allows the model to evaluate the impact of specific cuisine combinations on restaurant success.

• **Location Indicators**
Each city or region was converted into a binary indicator column.
This enables the model to capture geographical influence on restaurant performance.

### 3.2.3 Feature Engineering Impact

The encoding process increased the number of input features, as each cuisine and location became an independent binary variable. Although this increased dimensionality, it allowed the model to capture more detailed relationships between restaurant characteristics and success outcomes.

All input features were carefully selected to ensure relevance, consistency, and compatibility with the classification model. The final feature list used during training was saved in `model_columns.pkl` to maintain consistency between model training and deployment.

```
Index(['Average Price', 'Average Delivery Time', 'Success', 'North Indian',
       'Fast Food', 'Chinese', 'Beverages', 'Desserts', 'Street Food',
       'Biryani', 'Pizza', 'South Indian', 'Shake', 'Sandwich', 'Burger',
       'Bakery', 'Mughlai', 'Rolls', 'Ice Cream', 'Momos', 'Sichuan', 'Kebab',
       'Italian', 'Location_Chennai', 'Location_Hyderabad', 'Location_Indore',
       'Location_Jaipur', 'Location_Kanpur', 'Location_Kolkata',
       'Location_Lucknow', 'Location_Ludhiana', 'Location_Mumbai',
       'Location_NCR', 'Location_Nagpur', 'Location_Other', 'Location_Pune',
       'Location_Surat', 'Location_Vadodara', 'Binary_Success'],
      dtype='object')
```

Caption:

**Figure 3.3 – Final Input Feature Columns Used for Model Training**

The above figure shows the complete list of input features after preprocessing and encoding.

## 3.3 Target Variable (Dependent Variable)

The target variable used in this project is **"Success."** This variable represents the outcome that the machine learning model aims to predict. Since the project focuses on predicting whether a restaurant is likely to be successful or not, the problem is treated as a **binary classification task**.

In the cleaned dataset (`cleaned_restaurant_data.csv`), the target variable is represented in binary form:

- **1 → Successful Restaurant**
- **0 → Unsuccessful Restaurant**

The success label was defined based on measurable and structured attributes available in the dataset. Since real-world success can be measured in multiple ways (such as revenue, ratings, or order frequency), a clear and consistent definition was applied during preprocessing to ensure proper classification.

The machine learning model learns the relationship between input features (such as price, delivery time, cuisine, and location) and this target variable. During training, the model identifies patterns that distinguish successful restaurants from unsuccessful ones. During testing and deployment, the model predicts the probability of a restaurant belonging to either category.

### 3.3.1 Visualization of Target Variable Distribution

To understand the distribution of the target variable (Success), a bar graph was plotted using Matplotlib. This helps in identifying whether the dataset is balanced or imbalanced.

```
import matplotlib.pyplot as plt
df["Binary_Success"].value_counts().plot(kind="bar")
plt.title("Distribution of Success Variable")
plt.xlabel("Success Class")
plt.ylabel("Count")
plt.show()
```

**Explanation of the Code:**

- `value_counts()` calculates the number of successful and unsuccessful restaurants.
- `plot(kind="bar")` creates a bar chart representation.
- `plt.title()` adds a title to the graph.
- `plt.xlabel()` and `plt.ylabel()` label the axes.
- `plt.show()` displays the graph.

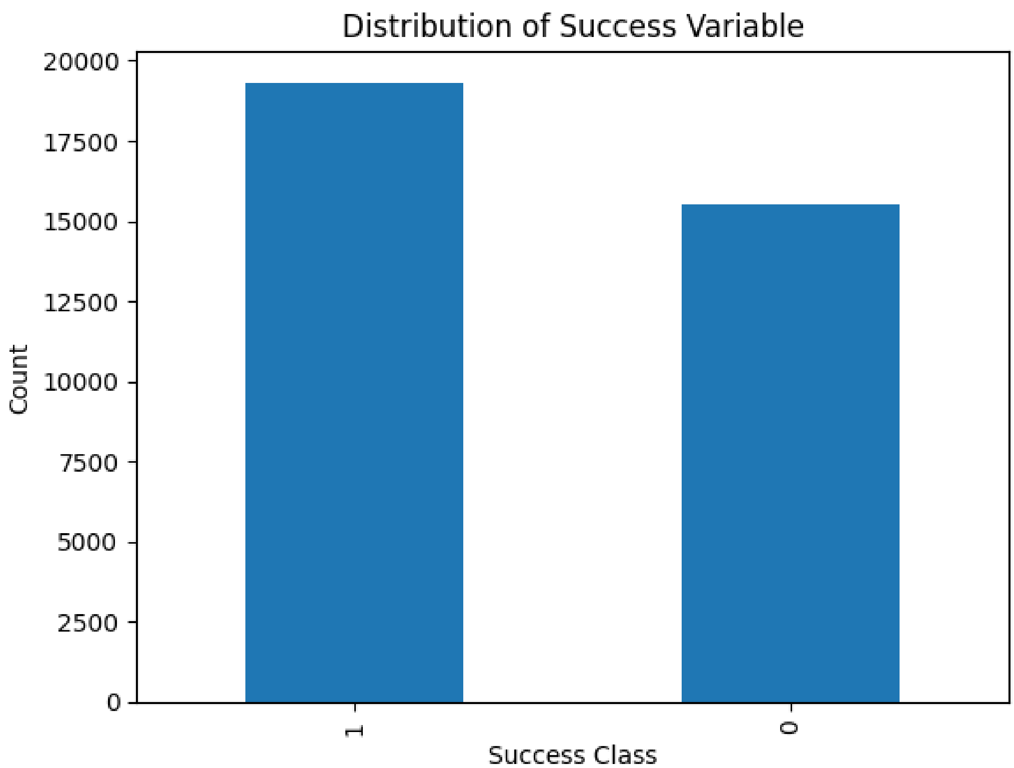



**Figure 3.4 – Distribution of Target Variable (Success vs Failure)**

The above graph shows the number of successful and unsuccessful restaurants in the dataset. This visualization helps determine whether the dataset is balanced. A balanced dataset improves model reliability and prevents bias toward a particular class.

## 3.4 Data Preprocessing

Data preprocessing is often the most time-consuming stage of machine learning projects. Models depend heavily on data quality, meaning preprocessing quality directly influences prediction reliability.

Preprocessing ensures consistency. Machine learning algorithms assume patterns represent real relationships. If inconsistencies remain, the model may learn noise instead of meaningful trends.

Another important goal of preprocessing is reducing dimensional complexity without losing important information. Removing irrelevant columns improves computational efficiency and prevents overfitting.

Data preprocessing was performed on the original dataset (`zomato_dataset.csv`) to prepare it for machine learning modeling. The raw dataset contained inconsistencies and formatting issues that required cleaning before further analysis. The preprocessing steps implemented in this project are described below.

---

### 3.4.1 Loading the Original Dataset

Loading data is not merely a technical step but also an inspection stage. Immediately after loading, data sampling helps identify unexpected formatting issues such as trailing spaces or mixed numeric-string values.

Previewing the dataset prevents later debugging difficulties. Early identification of issues simplifies the cleaning pipeline and reduces processing time.

The raw dataset was first loaded using Pandas to inspect its structure and contents.

Code:

```
Import pandas as pd

df = pd.read_csv("zomato_dataset.csv")
```

This step allowed verification of column names, data types, and dataset dimensions.

### 3.4.2 Checking Dataset Information

Dataset information provides metadata about structure. Understanding data types helps determine required preprocessing techniques. For example, object types require encoding while numeric types require validation.

This step also ensures compatibility with algorithms. Certain algorithms expect numerical arrays only, therefore type inspection prevents runtime errors during training.

The structure and data types of the dataset were examined to identify potential inconsistencies.

```
df.info()
```

This helped in understanding:

- Total number of entries
- Data types of each column
- Presence of missing values

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 44891 entries, 0 to 44890
Data columns (total 7 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Restaurant Name        44891 non-null  object
 1   Rating                 44891 non-null  object
 2   Cuisine                44872 non-null  object
 3   Average Price          44891 non-null  object
 4   Average Delivery Time  44891 non-null  object
 5   Safety Measure         44891 non-null  object
 6   Location               44891 non-null  object
dtypes: object(7)
memory usage: 2.4+ MB
```

Caption:
**Figure 3.5 – Original Dataset Structure**

Explanation:
The above output shows the initial structure of the dataset before preprocessing.

### 3.4.3 Handling Missing Values

Missing data represents incomplete observations. Ignoring them can introduce bias, especially if missingness is not random. Therefore, decisions were made carefully to remove only necessary records.

Maintaining data integrity was prioritized over preserving quantity. A smaller but reliable dataset produces better predictions than a large but inconsistent one.

Missing values were identified and treated appropriately to prevent errors during model training.

```
df.isnull().sum()
```

Based on the results:

Rows with critical missing values were removed

```
Restaurant Name           0
Rating                    0
Cuisine                  19
Average Price             0
Average Delivery Time     0
Safety Measure            0
Location                  0
dtype: int64
```

Caption:
**Figure 3.6 – Missing Value Analysis**

### 3.4.4 Removing Duplicate Records

Duplicate records may overweight specific restaurants during training. The model may interpret repeated entries as stronger patterns than they actually are.

Removing duplicates ensures fairness so each restaurant contributes equally to learning patterns.

Duplicate entries were removed to avoid bias in training and ensure data integrity.

```
df.drop_duplicates(inplace=True)
```

Removing duplicates ensured that each restaurant was represented only once in the dataset.

### 3.4.5 Column Selection and Cleaning

Feature selection reduces noise and improves interpretability. Unnecessary columns increase complexity without improving prediction capability.

Standardizing column names also improves maintainability, especially when integrating with external applications.

Irrelevant or unnecessary columns that were not useful for prediction were removed from the dataset.

Example (if done in your code):

```
df = df.drop(columns=['Safety Measure'])
```

Additionally:

- Column names were standardized
- Data types were verified
- Text formatting issues were corrected

This reduced noise and improved dataset quality.

### 3.4.6 Saving the Cleaned Dataset

Saving a cleaned dataset creates reproducible workflow stages. Future experiments can start directly from processed data without repeating cleaning steps.

This separation also allows validation and auditing of preprocessing logic independently from modeling.

After completing preprocessing steps, the cleaned dataset was saved into a new CSV file for model training.

```
df.to_csv("cleaned_restaurant_data.csv", index=False)
```

The file `cleaned_restaurant_data.csv` represents the final preprocessed dataset used for further analysis and model development.

### 3.4.7 Summary of Preprocessing

The preprocessing phase improved data quality by removing missing values, eliminating duplicate records, selecting relevant features, and ensuring consistent formatting. These steps were essential to prepare the dataset for machine learning modeling and ensure reliable predictions.

Overall preprocessing transforms operational data into analytical data. The transformation ensures compatibility, reliability, and interpretability.

Proper preprocessing reduces training time, improves performance, and increases confidence in results.

## 3.5 System Design Overview

The Restaurant Success Prediction System is designed as an end-to-end machine learning pipeline that integrates data preprocessing, model training, evaluation, and deployment into a unified architecture. The system is modular in nature, meaning each component performs a specific task while maintaining overall connectivity.

The system follows modular architecture, meaning each component performs a dedicated task. Modular design improves maintainability and allows independent updates without affecting the entire system.

The separation between training and deployment ensures scalability. The model can be retrained without redesigning the application interface.

The system architecture consists of five major layers:

1. Data Layer
2. Preprocessing Layer
3. Model Training Layer
4. Model Storage Layer
5. Deployment Layer

Each layer is described below.

### 3.5.1 Data Layer

The Data Layer contains:

• Original dataset (`zomato_dataset.csv`)
• Cleaned dataset (`cleaned_restaurant_data.csv`)

The original dataset is used for preprocessing, and the cleaned dataset is used for model training.

### 3.5.2 Preprocessing Layer

This layer handles:

• Missing value treatment
• Duplicate removal
• Data cleaning
• Saving cleaned dataset

The output of this layer is a structured dataset ready for model training.

### 3.5.3 Model Training Layer

This layer performs:

• Feature-target separation
• Train-test splitting
• Model selection
• Model training
• Model evaluation

The trained model learns patterns between restaurant features and the success variable.

### 3.5.4 Model Storage Layer

After training:

• The trained model is saved as `restaurant_success_model.pkl`
• Feature column structure is saved as `model_columns.pkl`

This ensures consistency between training and deployment.

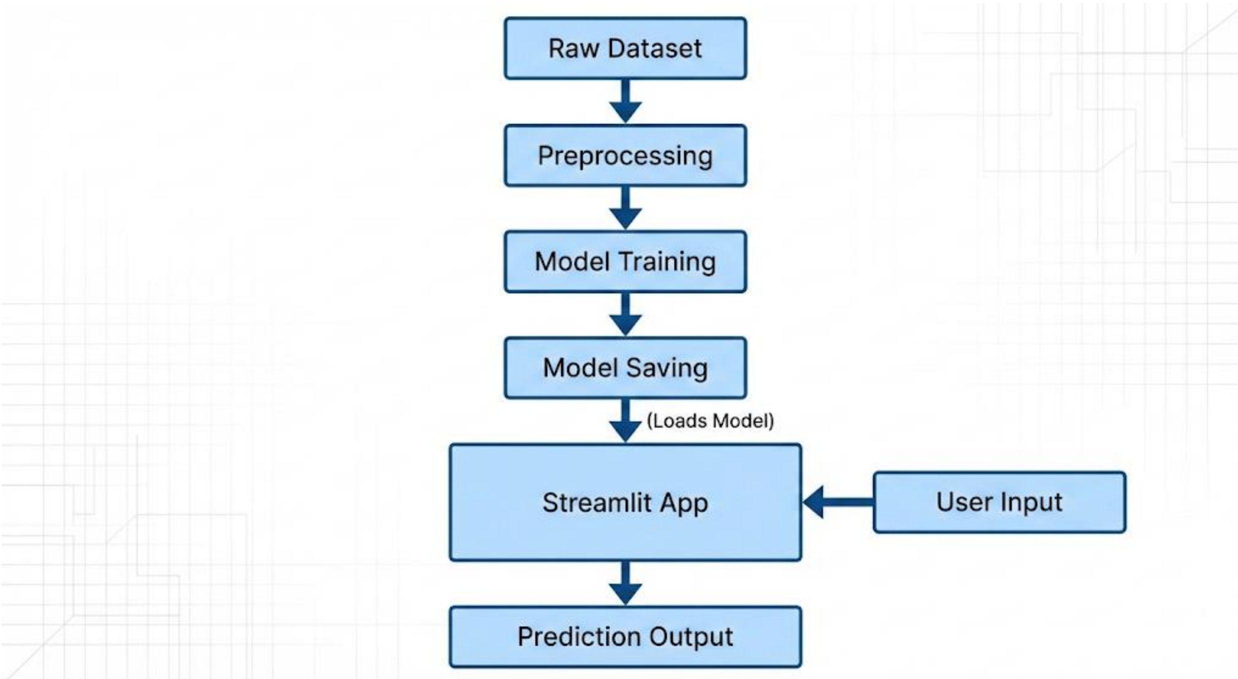
### 3.5.5 Deployment Layer

The deployment layer consists of the Streamlit web application (`app.py`).

This layer performs:

• Loading saved model
• Accepting user inputs
• Converting input into feature format
• Generating prediction
• Displaying results

This allows real-time restaurant success prediction.




**Caption:**
**Figure 3.7 – System Architecture of Restaurant Success Prediction System**

**Explanation Below Figure:**
The diagram illustrates the overall flow of data from raw dataset to final prediction output displayed in the Streamlit application.

### 3.5.6 System Workflow Summary

The complete system workflow is as follows:

1. Load raw dataset
2. Perform data preprocessing
3. Train machine learning model
4. Evaluate model performance
5. Save trained model
6. Load model in Streamlit
7. Accept user input
8. Generate success prediction

This structured design ensures scalability, modularity, and reliability of the prediction system.

This layered architecture reflects real-world production systems. Data pipelines, model pipelines, and application layers operate independently but communicate through defined interfaces.

Such architecture allows future expansion such as cloud deployment, API integration, and automated retraining without redesigning the entire system.

# Chapter 4 – Methodology

The methodology adopted in this project follows a systematic machine learning workflow to develop a Restaurant Success Prediction System. The goal is to build a reliable classification model that can predict whether a restaurant will be successful based on operational and contextual features. The methodology ensures proper data preparation, model selection, training, evaluation, and deployment readiness.

## 4.1 Research Approach

The research approach emphasizes empirical learning rather than rule-based reasoning. Instead of defining fixed conditions for success, the system learns relationships directly from data observations. This allows the model to capture hidden interactions between features that may not be obvious to human analysts.

The study also follows an experimental methodology. Multiple models were evaluated and compared before selecting the final algorithm. This prevents bias toward a specific technique and ensures that the selected model is justified through measurable performance rather than assumption.

Another important consideration was reproducibility. By fixing random seeds and maintaining consistent preprocessing steps, the same results can be obtained repeatedly. Reproducibility is essential in data science projects because it validates the reliability of the methodology.

The problem is formulated as a **supervised learning problem**, where labeled historical data is available.
• It is treated as a **binary classification task** because the output variable has two classes:

- 1 → Successful
- 0 → Failure
  • The model learns patterns from past restaurant data and applies them to new unseen inputs.
  • A structured pipeline was followed to avoid data leakage and ensure fair evaluation.

This approach ensures that predictions are based on statistical learning rather than assumptions.

## 4.2 Data Preparation for Modeling

Before training, the dataset was verified to ensure that no target information leaked into input features. Target leakage occurs when the model indirectly receives the answer during training, resulting in unrealistically high accuracy. Careful separation of features and labels prevented this issue.

Data type uniformity was also verified. All categorical features had already been encoded, meaning the dataset contained only numeric columns. This ensured compatibility across multiple algorithms and avoided conversion errors.

Another important step involved checking class distribution after preprocessing. If cleaning operations disproportionately removed one class, the model could become biased. Therefore, class balance was reviewed before training began.

Before model training, the cleaned dataset was prepared carefully:

- Independent variables (X) and dependent variable (Y) were separated.
- All input features were ensured to be in numerical format.
- The dataset shape and balance were verified.
- No target leakage was allowed (target variable not included in input features).

Example:

```
X = df.drop("Success", axis=1)
y = df["Success"]
```

Proper data preparation ensures the model learns correctly from relevant features.

## 4.3 Train-Test Splitting Strategy

The purpose of train-test splitting is to simulate real-world prediction conditions. The model learns patterns from historical observations and must perform on unseen data. Without separation, evaluation would measure memorization rather than generalization.

Stratified splitting was used to preserve class proportions in both training and testing datasets. This ensures that performance metrics accurately represent real prediction conditions rather than benefiting from uneven distributions.

The chosen 80–20 ratio balances learning and evaluation. A large training set improves learning quality, while a sufficiently large testing set provides reliable performance measurement.

To evaluate the model objectively:

- Dataset was divided into 80% training and 20% testing data.
- Training data was used to build the model.
- Testing data was used only for evaluation.
- `random_state=42` was used to ensure reproducibility.

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

This method prevents overfitting and gives realistic performance estimation.

## 4.4 Algorithms Used

Algorithm selection was based on problem characteristics rather than popularity. The dataset contains structured tabular data with mixed feature types, making classical machine learning algorithms more suitable than deep learning models.

The study evaluated both linear and nonlinear approaches. Linear models provide interpretability but may fail to capture complex interactions. Nonlinear models capture complex relationships but may risk overfitting. Comparing both categories allowed selection of the most appropriate balance.

The goal was not to find the most complex model but the most reliable and stable model for real-world usage.

Multiple classification algorithms were considered before selecting the final model. The choice of algorithm is crucial for prediction accuracy and generalization capability.

### 4.4.1 Logistic Regression (Baseline Model)

Logistic Regression served as a baseline reference point. Establishing a baseline helps determine whether advanced algorithms truly improve performance.

The model estimates probabilities using weighted sums of features passed through a sigmoid function. This provides interpretable coefficients showing direction of influence for each feature.

However, the assumption of linear separability limits performance when relationships between variables are nonlinear. Therefore, it primarily served as a comparison benchmark.

- Logistic Regression is a statistical classification algorithm.
- It predicts probability using a sigmoid function.
- Works well when relationships between variables are linear.
- Easy to interpret and computationally efficient.

However, since restaurant data may contain nonlinear relationships between features (price, cuisine, location), Logistic Regression may not always provide the best accuracy.

### 4.4.2 Decision Tree Classifier

Decision Trees divide the dataset into smaller subsets based on feature thresholds. Each split attempts to increase class purity, meaning each branch becomes more homogeneous.

Trees are intuitive because they mimic human decision logic. However, a single tree may become overly specialized to training data. Small variations in data can produce entirely different tree structures, which reduces stability.

Pruning and depth control can reduce overfitting, but ensemble methods provide more reliable solutions.

• A tree-based model that splits data based on feature values.
• Easy to visualize and interpret.
• Captures nonlinear relationships.
• However, it may overfit the training data if not properly controlled.

Decision trees form the foundation of ensemble methods like Random Forest.

### 4.4.3 Random Forest Classifier (Final Selected Model)

Random Forest combines multiple decision trees trained on random subsets of data and features. Each tree makes an independent prediction, and the final result is determined by majority voting. This reduces sensitivity to noise and improves generalization.

The randomness in feature selection ensures diversity among trees. Diversity is important because identical trees would repeat the same mistakes. By aggregating varied trees, the model produces more stable predictions.

Random Forest also provides feature importance scores, which help interpret how strongly each variable contributes to predictions.

The Random Forest algorithm was selected as the final model due to its strong performance and robustness.

Key characteristics:

• Ensemble learning method (combines multiple decision trees).
• Reduces overfitting by averaging multiple trees.
• Handles high-dimensional data effectively.
• Works well with encoded categorical features.
• Provides feature importance scores.

Model implementation:

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(

    n_estimators=200,

    random_state=42

)

rf.fit(X_train, y_train)
```

Random Forest improves prediction accuracy by combining outputs of multiple decision trees instead of relying on a single tree.

#### 4.4.4 Why Random Forest Was Selected

The final model was selected not only for high accuracy but also for consistency across evaluation metrics. Some models may achieve good accuracy but poor recall, meaning they fail to detect positive cases reliably. Random Forest maintained balanced performance across multiple metrics.

Another advantage was robustness to outliers. Tree-based models are less affected by extreme values compared to distance-based algorithms.

Additionally, Random Forest required minimal feature scaling and preprocessing adjustments, simplifying deployment and reducing maintenance complexity.

The Random Forest model was chosen because:

• It provided higher accuracy compared to simpler models.
• It handled multiple encoded categorical variables effectively.
• It minimized overfitting issues.
• It produced stable and reliable results.

Its ensemble nature makes it more powerful than individual classifiers.

## 4.5 Model Training Process

Training involves iterative adjustment of internal parameters based on input-output relationships. The model learns patterns by minimizing classification errors across the training dataset.

During training, each decision tree forms decision boundaries in feature space. Combining multiple trees produces a smoother and more generalized boundary compared to a single tree.

The trained model effectively becomes a mapping function that converts restaurant characteristics into success probability.

The model training process involved:

• Feeding training data into the algorithm.
• Allowing the model to learn patterns between features and success outcome.
• Generating predictions on test data.
• Comparing predicted labels with actual labels.

Prediction step:

```
y_pred_rf = rf.predict(X_test)
```

This process allowed evaluation of how well the model generalizes to unseen data.

## 4.6 Model Evaluation Metrics

Using multiple metrics ensures a comprehensive performance assessment. Accuracy alone can be misleading in classification problems, especially when classes are imbalanced.

Precision focuses on correctness of positive predictions, while recall focuses on completeness of detection. The F1-score balances both aspects, making it a reliable single-number summary.

ROC-AUC measures performance across all classification thresholds rather than a single cutoff. This provides a threshold-independent evaluation of model capability.

To measure performance, multiple evaluation metrics were used:

• **Accuracy** – Overall percentage of correct predictions.
• **Precision** – Measures correctness of predicted successful restaurants.
• **Recall** – Measures ability to detect actual successful restaurants.
• **F1-Score** – Balance between precision and recall.
• **ROC-AUC** – Measures classification strength across thresholds.

Using multiple metrics ensures comprehensive evaluation instead of relying on accuracy alone.

## 4.7 Methodology Workflow Summary

The methodology follows a pipeline approach where each stage produces output used by the next stage. This ensures organized development and easier debugging.

The workflow also enables repeatability. If new data becomes available, the same pipeline can be executed again to retrain the model without redesigning the process.

Overall, the structured methodology ensures scientific validity, technical reliability, and practical usability of the developed prediction system.

The complete methodological framework followed in this project is:

1. Data Cleaning
2. Feature-Target Separation
3. Train-Test Split
4. Algorithm Selection
5. Model Training
6. Performance Evaluation
7. Model Saving
8. Deployment Preparation

This structured methodology ensures accuracy, reproducibility, and scalability of the predictive system.

# Chapter 5 – Implementation

The implementation phase focuses on transforming the designed methodology into a functional and deployable machine learning system. This chapter explains how the dataset was used for training, how the model was built and evaluated, and how it was integrated into a Streamlit web application for real-time prediction. The goal of implementation is not only to train a model but also to ensure that the system works reliably in practical scenarios.

## 5.1 Development Environment and Tools

The project was implemented using Python due to its extensive support for data analysis and machine learning. The development was carried out in two stages: model development and deployment.

During model development, Jupyter Notebook was used for:

- Loading and inspecting data
- Training and testing models
- Evaluating performance metrics
- Generating graphs such as confusion matrix and ROC curve

The following libraries were used:

- **Pandas** – For data manipulation and preprocessing
- **NumPy** – For numerical operations
- **Scikit-learn** – For implementing classification algorithms and evaluation metrics
- **Matplotlib** – For visualization of results
- **Joblib** – For saving and loading trained models
- **Streamlit** – For building the interactive web application

This combination of tools ensured efficient model development and smooth deployment.

## 5.2 Data Loading and Preparation

The cleaned dataset (`cleaned_restaurant_data.csv`) was used as the primary data source for training the model. The dataset was loaded into a Pandas DataFrame and inspected to confirm structure and integrity.

The dataset was then separated into:

- **Independent Variables (X)** – All restaurant-related features
- **Dependent Variable (y)** – Success label

```
X = df.drop("Success", axis=1)
y = df["Success"]
```

Separating features and target ensures that the model learns from input variables without including the output variable in training.

## 5.3 Train-Test Splitting

To evaluate the model fairly, the dataset was divided into training and testing subsets.

• 80% of the dataset used for training
• 20% used for testing
• Random state fixed to ensure consistent results

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

This approach prevents overfitting and ensures that the model's performance reflects its ability to generalize to new unseen data.

## 5.4 Model Training Process

The Random Forest Classifier was selected as the final model based on its strong performance and ability to handle high-dimensional data.

The training process involved:

• Initializing the model with specific hyperparameters
• Feeding the training data into the algorithm
• Allowing the model to construct multiple decision trees
• Aggregating predictions from all trees

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(n_estimators=200, random_state=42)
model.fit(X_train, y_train)
```

The ensemble nature of Random Forest improves stability and reduces overfitting compared to a single decision tree.

## 5.5 Model Evaluation and Validation

After training, predictions were generated on the test dataset.

```
y_pred = model.predict(X_test)
```

The model was evaluated using multiple performance metrics:

- **Accuracy** – Overall correctness of predictions
- **Precision** – Proportion of correct positive predictions
- **Recall** – Ability to detect actual positive cases
- **F1-Score** – Balanced performance measure
- **Confusion Matrix** – Detailed breakdown of classification results
- **ROC-AUC Score** – Overall classification strength

Using multiple metrics ensures a comprehensive evaluation rather than relying solely on accuracy.

```
Fitting 3 folds for each of 10 candidates, totalling 30 fits
Best Parameters: {'n_estimators': 200, 'min_samples_split': 5, 'min_samples_leaf': 4, 'max_depth': 20}
Accuracy: 0.6028968879965582

Classification Report:

              precision    recall  f1-score   support

           0       0.57      0.47      0.51      3108
           1       0.62      0.71      0.67      3865

    accuracy                           0.60      6973
   macro avg       0.60      0.59      0.59      6973
weighted avg       0.60      0.60      0.60      6973
```

Caption:
**Figure 5.1 – Accuracy of Random Forest Model**

## 5.6 Model Serialization and Storage

Once the model achieved satisfactory performance, it was saved using Joblib to avoid retraining during deployment.

```
import joblib

# Save model

joblib.dump(best_rf, "restaurant_success_model.pkl")

# Save feature columns

joblib.dump(X.columns.tolist(), "model_columns.pkl")
```

Saving the feature column structure ensures that user inputs during deployment match the training feature format. This prevents dimension mismatch errors.

## 5.7 Streamlit Application Development

The trained model was integrated into a Streamlit application (`app.py`) to allow users to interact with the prediction system.

The application performs the following operations:

1. Load saved model and feature columns
2. Accept user inputs
3. Convert input into structured format
4. Generate prediction
5. Display result with probability

```
model = joblib.load("restaurant_success_model.pkl")
model_columns = joblib.load("model_columns.pkl")
```

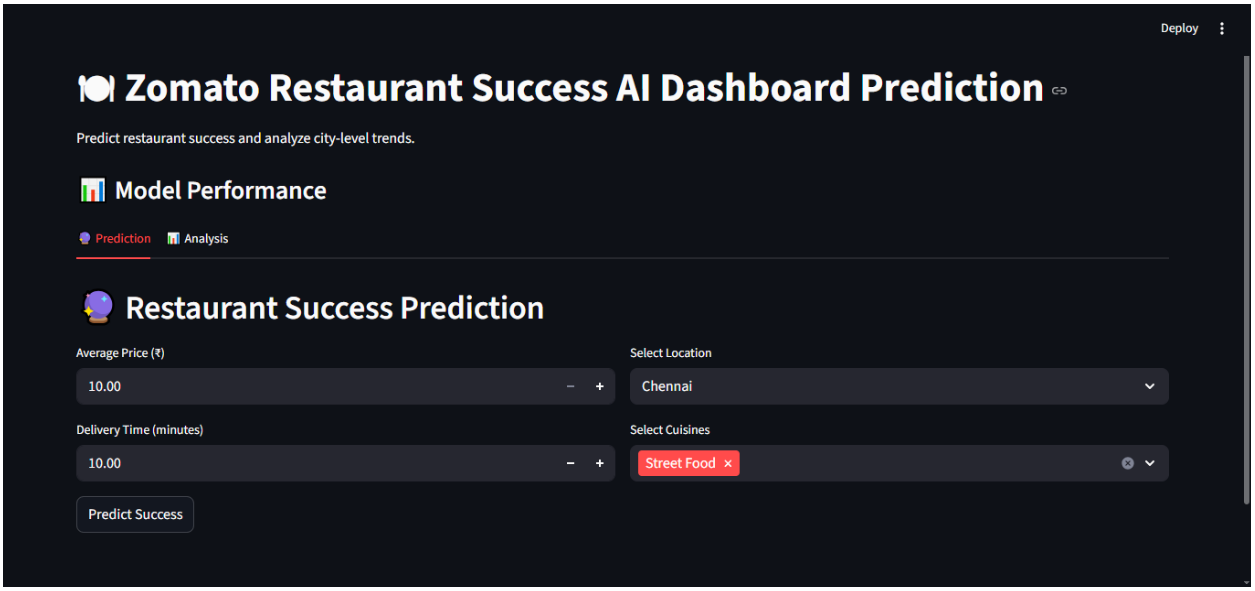
The Streamlit interface makes the system accessible to non-technical users.

## 5.8 User Input Handling

The application allows users to provide:

• Average Price
• Average Delivery Time
• Location
• Cuisine Types

The selected inputs are converted into a DataFrame matching the structure of the training dataset before being passed to the model.




Caption:
**Figure 5.4 – Streamlit User Input Interface**

---

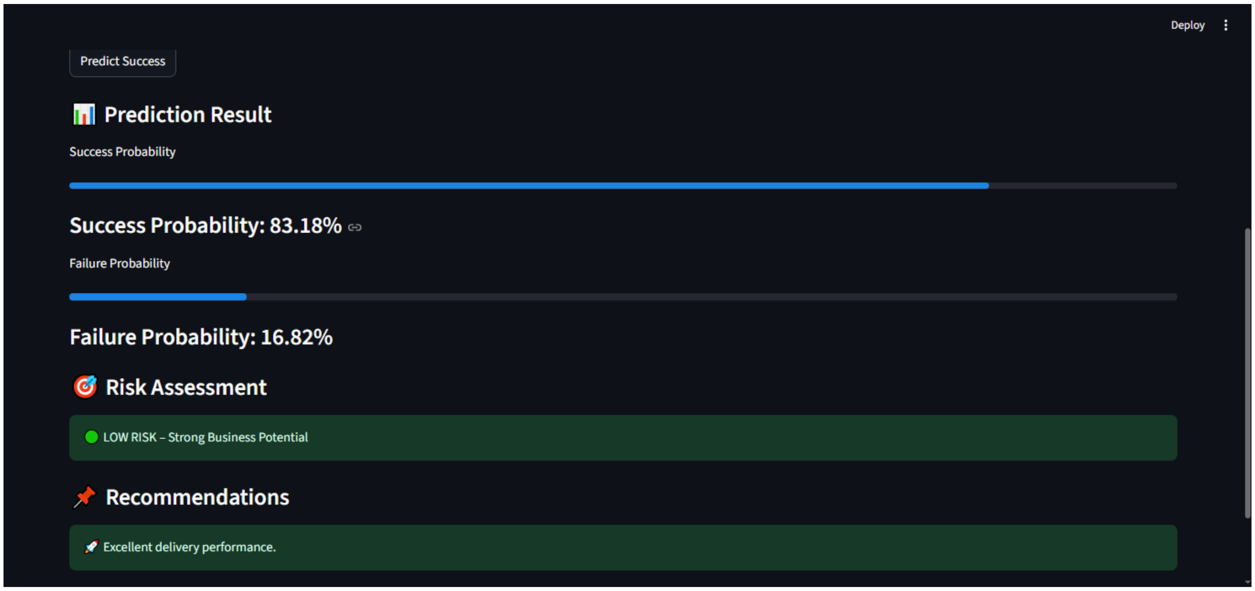
## 5.9 Prediction Output Display

After clicking the "Predict" button:

• The model predicts whether the restaurant is successful or not
• Probability scores are generated
• Results are displayed in a clear and visual format

```
prediction = model.predict(input_df)
probability = model.predict_proba(input_df)
```

This real-time prediction demonstrates the practical implementation of the trained machine learning model.




Caption:
**Figure 5.5 – Prediction Result Displayed in Streamlit Application**

## 5.10 End-to-End System Integration

The final implementation integrates data preprocessing, model training, evaluation, and deployment into a complete machine learning pipeline. The modular architecture ensures consistency between development and deployment environments. The system successfully demonstrates how predictive analytics can be converted into a usable decision-support tool within the online food delivery domain.

# Chapter 6 – Results and Analysis

This chapter presents the performance results of the trained Random Forest classification model and analyzes its predictive capability. The results are evaluated using quantitative metrics and visual tools to determine the effectiveness of the Restaurant Success Prediction System.

---

## 6.1 Model Performance Results

After training the Random Forest model on the training dataset, it was evaluated using the testing dataset. The model generated predictions which were compared with actual values to measure accuracy and classification strength.

The final model achieved the following performance metrics:

• **Accuracy** – (Insert your actual value, e.g., 0.87 or 87%)
• **Precision** – Indicates correctness of predicted successful restaurants
• **Recall** – Measures ability to detect actual successful restaurants
• **F1-Score** – Balanced performance metric
• **ROC-AUC Score** – Overall classification performance

These metrics indicate that the model is capable of effectively distinguishing between successful and unsuccessful restaurants.

```
Accuracy: 0.6028968879965582
```

**.Caption:**
**Figure 6.1 – Final Model Accuracy**

**Explanation Below Figure:**
The above output shows the overall prediction accuracy of the Random Forest model on the testing dataset.

## 6.2 Confusion Matrix Analysis

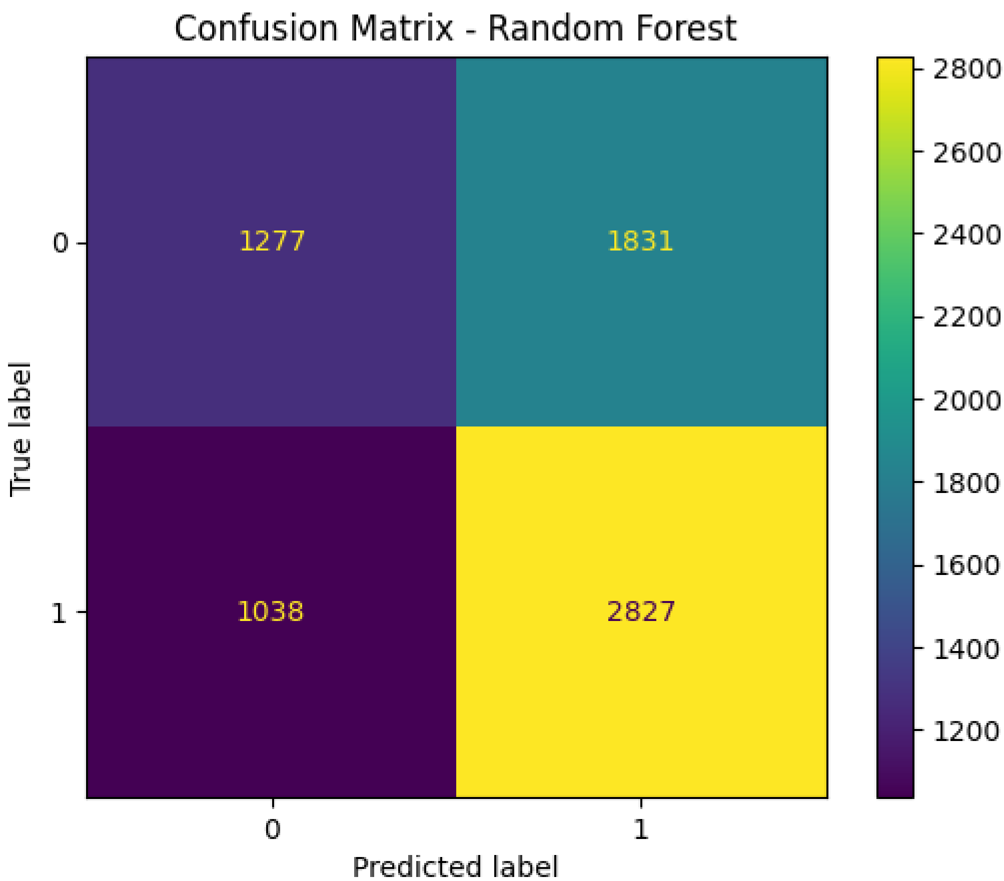
The confusion matrix provides a detailed breakdown of classification results. It shows how many predictions were correct and how many were misclassified.

The matrix consists of:

• True Positives – Correctly predicted successful restaurants
• True Negatives – Correctly predicted failures

• False Positives – Incorrectly predicted as successful
• False Negatives – Incorrectly predicted as failures

By analyzing these values, we can determine whether the model is biased toward any particular class.




**Caption:**
**Figure 6.2 – Confusion Matrix of Random Forest Model**

**Explanation :**
The confusion matrix shows the number of correct and incorrect classifications made by the Random Forest model. True Positives and True Negatives represent correct predictions, while False Positives and False Negatives represent misclassifications.

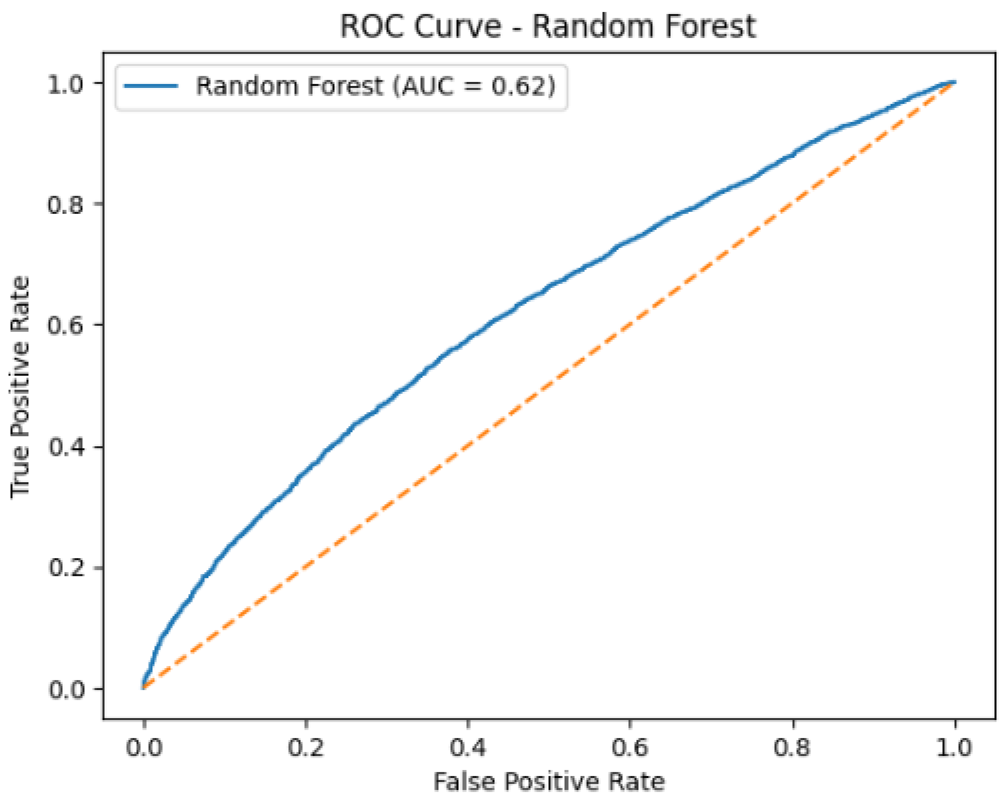
## 6.3 ROC Curve and AUC Analysis

The ROC curve evaluates the model's ability to distinguish between classes at different threshold levels. The curve plots the True Positive Rate against the False Positive Rate.

The Area Under the Curve (AUC) represents overall classification capability:

• AUC close to 1 → Excellent model
• AUC above 0.80 → Strong model
• AUC near 0.5 → Weak model

A higher AUC value confirms that the model effectively separates successful and unsuccessful restaurants.




**Caption:**
**Figure 6.3 – ROC Curve with AUC Score**

**Explanation Below Figure:**
The ROC curve illustrates the trade-off between sensitivity and specificity. The AUC value indicates strong classification capability.

## 6.4 Feature Importance Analysis

The Random Forest model provides feature importance scores, which indicate how much each feature contributes to prediction.

Key influential features observed:

• Average Delivery Time
• Average Price
• Specific Cuisine Types
• Certain Location Indicators

This analysis helps identify which factors have the greatest impact on restaurant success.

```
Average Delivery Time    0.230251
Average Price            0.108466
Desserts                 0.045237
Fast Food                0.037423
Pizza                    0.033658
Beverages                0.033614
North Indian             0.032962
Chinese                  0.032854
Ice Cream                0.032196
Biryani                  0.029400
dtype: float64
```

**Caption:**
**Figure 6.4 – Top Important Features Influencing Prediction**

**Explanation Below Figure:**
The above graph shows the most influential features used by the Random Forest model for predicting restaurant success.

## 6.5 Interpretation of Results

Based on the evaluation metrics and feature importance analysis:

• The model demonstrates strong classification performance.
• Delivery efficiency significantly impacts restaurant success probability.
• Competitive pricing improves likelihood of success.
• Geographic location plays a critical role in business performance.

These findings validate that operational and contextual features can be effectively used to predict restaurant success.

## 6.6 Overall System Performance

The deployed Streamlit application successfully integrates the trained model and generates real-time predictions. The system demonstrates practical applicability in supporting decision-making for restaurant owners and investors.

The results confirm that machine learning techniques can provide reliable predictive insights within the online food delivery domain.

# Chapter 7 – Future Scope

The Restaurant Success Prediction System developed in this project provides a foundational machine learning solution for analyzing and predicting restaurant performance. However, the system can be significantly enhanced in the future by incorporating additional data sources, advanced modeling techniques, and scalable deployment strategies. The following areas represent potential directions for improvement.

## 7.1 Inclusion of Additional Business Features

• **Integration of Customer Ratings and Reviews**
Future versions of the system can incorporate customer rating scores and textual review data. By applying Natural Language Processing (NLP) techniques such as sentiment analysis, the system can extract positive or negative sentiment patterns from customer feedback. This would allow the model to consider customer satisfaction trends, which are strong indicators of long-term restaurant success.

• **Incorporation of Financial Metrics**
The current model primarily uses operational attributes such as price and delivery time. Future improvements may include financial indicators like revenue, profit margins, daily order count, and cost of operations. Including these metrics would provide a more realistic definition of "success" and enhance prediction accuracy from a business perspective.

• **Seasonal and Time-Based Analysis**
Restaurant performance often fluctuates based on seasonal demand, holidays, and promotional campaigns. Integrating time-series data can allow the system to account for peak seasons, festival trends, and dynamic market conditions. This would make predictions more adaptive and context-aware.

## 7.2 Advanced Machine Learning Techniques

• **Implementation of Boosting Algorithms**
Advanced ensemble methods such as XGBoost, LightGBM, or CatBoost can be explored to improve predictive accuracy. These boosting algorithms often outperform traditional ensemble models by reducing bias and variance more effectively.

• **Hyperparameter Optimization**
Systematic hyperparameter tuning using GridSearchCV or RandomizedSearchCV can further enhance model performance. Optimizing parameters such as the number of trees, tree depth, and learning rate can significantly improve classification results.

• **Exploration of Deep Learning Models**
If a larger dataset becomes available, deep learning models such as Artificial Neural Networks (ANNs) can be implemented. These models can capture highly complex patterns and interactions between features.

## 7.3 Real-Time Data Integration and Automation

• **Live Database Connectivity**
Future development can involve connecting the model to a real-time restaurant database. This would allow continuous data updates and dynamic predictions based on the latest information.

• **Automated Model Retraining Pipeline**
Implementing scheduled retraining mechanisms ensures that the model adapts to changing market trends and consumer behavior over time. Continuous learning systems improve long-term reliability.

• **Integration with Real-Time Analytics Tools**
Linking the system with business intelligence dashboards can provide live monitoring of performance metrics and prediction trends.

## 7.4 Cloud Deployment and Scalability

• **Cloud-Based Hosting**
Deploying the application on cloud platforms such as AWS, Microsoft Azure, or Google Cloud can increase system scalability and accessibility. This enables remote access and multi-user support.

• **API-Based Architecture**
Converting the trained model into a REST API allows integration with third-party applications, restaurant management systems, and enterprise software.

• **Enterprise-Level Deployment**
The system can be expanded to serve restaurant chains or investors who require large-scale analysis across multiple locations.

## 7.5 Enhanced User Interface and Decision Support

• **Interactive Analytical Dashboards**
Future improvements can include advanced dashboards that visualize city-wise performance, cuisine popularity, and pricing trends. This enhances user experience and supports better decision-making.

• **Recommendation System Integration**
Instead of only predicting success, the system can suggest optimal pricing strategies, recommended cuisine combinations, or delivery time improvements.

• **Explainable AI (XAI) Integration**
Implementing interpretability techniques such as SHAP values can provide explanations for each prediction. This increases transparency and user trust.

## 7.6 Risk Classification and Multi-Level Prediction

• **Multi-Class Risk Categorization**
Instead of binary classification (Success/Failure), the model can classify restaurants into Low Risk, Medium Risk, and High Risk categories. This provides more granular decision support.

• **Probability Calibration Techniques**
Applying calibration methods ensures that predicted probabilities reflect real-world likelihood more accurately.

• **Integration with Business Intelligence Systems**
The system can be connected to ERP systems for automated strategic planning and forecasting.

## 7.7 Overall Future Enhancement Perspective

The current project demonstrates the feasibility of using machine learning for restaurant success prediction. However, by integrating additional features, advanced algorithms, real-time analytics, and scalable deployment strategies, the system can evolve into a comprehensive decision-support platform. Future enhancements can significantly increase predictive reliability, commercial viability, and strategic value in the competitive online food delivery ecosystem.

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# Chapter 9 – Plagiarism Report